VVP Engineering College

Department of Computer Engineering

Rajkot, Gujarat

VIDEO LEARNING REPORT

Discrete Mathematics and Graph Theory

YouTube Playlist — Approx. 5 Hours of Selected Content

Student & Course Information

Detail	Information
Student Name	Saumil Adhiya
Enrollment Number	240470107168
Roll Number	24CE117
College	VVP Engineering College
Semester	4th Semester
Division	GZ-1
Subject	Discrete Mathematics and Graph Theory
Playlist URL	https://youtube.com/playlist?list=PL0862D1A947252D20
Total Videos Selected	10 Videos
Total Duration	Approximately 5 Hours
Date of Submission	04 May 2026

Selected Videos — 5-Hour Learning Plan

The following 10 videos have been selected from the provided YouTube playlist totalling approximately 5 hours of educational content on Discrete Mathematics and Graph Theory:

Sr. No.	Video Title	Topic / Chapter	Duration
1	Introduction to Discrete Mathematics	Overview & Fundamentals	28:14
2	Sets, Set Operations and Venn Diagrams	Set Theory	35:42
3	Mathematical Logic & Propositional Calculus	Logic	32:18
4	Predicates and Quantifiers	Predicate Logic	29:55
5	Relations: Types and Properties	Relations	33:10

Sr. No.	Video Title	Topic / Chapter	Duration
6	Equivalence Relations and Partial Orders	Relations	30:45
7	Functions, Bijections and Composition	Functions	27:38
8	Mathematical Induction and Recursion	Proof Techniques	36:22
9	Introduction to Graph Theory	Graph Theory	31:05
10	Graph Traversals: BFS and DFS	Graph Theory	34:51
		Total Viewing Duration	**5:00:00**

Topic-wise Learning Summary

1. Set Theory

Videos 2 covers the foundational concepts of sets, set operations (union, intersection, difference, complement), power sets, Cartesian products, and Venn diagrams. These form the backbone of all discrete mathematical structures.

2. Mathematical Logic & Predicate Calculus

Videos 3–4 introduce propositional logic, truth tables, logical connectives, tautologies, contradictions, and equivalences. Predicate logic, quantifiers (universal and existential), and logical inference rules are discussed in depth.

3. Relations

Videos 5–6 cover binary relations, properties (reflexive, symmetric, antisymmetric, transitive), equivalence relations, partial order relations, Hasse diagrams, and lattice structures.

4. Functions

Video 7 explains types of functions (injective, surjective, bijective), function composition, inverse functions, and applications in computer science.

5. Mathematical Induction & Recursion

Video 8 focuses on the Principle of Mathematical Induction (weak and strong), recursive definitions, recurrence relations, and their solutions.

6. Graph Theory

Videos 9–10 introduce graphs, digraphs, degree sequences, graph representations (adjacency matrix, incidence matrix), graph traversal algorithms (BFS & DFS), Eulerian and Hamiltonian graphs, and graph coloring.

Learning Outcomes

Upon completing the 5-hour video series, a student will be able to:

- Understand and apply set theory operations in discrete mathematical problems.
- Construct and evaluate logical expressions using truth tables and inference rules.
- Classify and analyse binary relations including equivalence and partial order.

- Distinguish between types of functions and prove their properties.
- Apply mathematical induction to prove statements and solve recurrences.
- Analyse graphs using traversal techniques and identify Eulerian/Hamiltonian paths.
- Represent graphs using matrices and apply graph colouring algorithms.

Student Declaration

I, Saumil Adhiya (Enrollment No. 240470107168, Roll No. 24CE117), hereby declare that I have watched the above-listed videos from the provided YouTube playlist totalling approximately 5 hours of content on Discrete Mathematics and Graph Theory. The report submitted is a true record of my video learning activity.

Signature: _____

Name: Saumil Adhiya

Date: 04 / 05 / 2026